

Assignment 3: Sentiment Analysis using Naive Bayes

Aim

To develop a machine learning program that recognizes whether a document or sentence is Positive or Negative using polarity words and a suitable classification algorithm.

Objectives

- Understand sentiment analysis and text classification.
- Learn text preprocessing techniques.
- Convert text into numerical form using TF-IDF.
- Train a Naive Bayes classification model.
- Predict Positive or Negative sentiment from user input.
- Visualize model performance using graphs.

Theory

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone of a text. It classifies text into categories such as Positive or Negative.

In this practical, the dataset contains sentences with corresponding sentiments. The text is converted into numerical form using TF-IDF Vectorization and classified using the Multinomial Naive Bayes algorithm.

Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- TF-IDF Vectorizer
- Multinomial Naive Bayes

Important Formulas

1. TF-IDF Formula

$$\text{TF-IDF} = \text{TF} \times \text{IDF}$$

$$\text{TF} = (\text{Number of times word appears in document}) / (\text{Total words in document})$$

$$\text{IDF} = \log(\text{Total Documents} / \text{Documents containing word})$$

2. Accuracy Formula

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

Algorithm Used - Naive Bayes

Multinomial Naive Bayes is a supervised machine learning algorithm commonly used for text classification problems.

Advantages:

- Fast and efficient
- Works well for text data
- Easy to implement
- Suitable for sentiment analysis

Code Explanation

Step 1: Import required libraries.

Step 2: Create dataset containing positive and negative sentences.

Step 3: Display sentiment distribution graph.

Step 4: Convert text into numerical vectors using TF-IDF.

Step 5: Split dataset into training and testing sets.

Step 6: Train Naive Bayes model.

Step 7: Predict sentiments and calculate accuracy.

Step 8: Display confusion matrix graph.

Step 9: Accept live user input and predict sentiment.

Visualizations Included

- Sentiment Distribution Count Plot
- Confusion Matrix Heatmap

Instructions to Run .py File in Anaconda

Step 1: Open Anaconda Prompt.

Step 2: Install required libraries:

```
pip install pandas matplotlib seaborn scikit-learn
```

Step 3: Navigate to the folder containing the Python file.

Step 4: Run the file using:

```
python filename.py
```

Instructions to Run .ipynb File in Jupyter Notebook

Step 1: Open Anaconda Navigator.

Step 2: Launch Jupyter Notebook.

Step 3: Open the .ipynb notebook file.

Step 4: Run each cell one by one using Shift + Enter.

Expected Output

- Positive and Negative sentiment predictions.
- Accuracy score.
- Confusion matrix graph.
- Sentiment distribution graph.
- Live sentiment prediction from user input.

Outcome

Successfully implemented a machine learning-based sentiment analysis system that classifies text as Positive or Negative using TF-IDF and Naive Bayes.

Conclusion

The practical demonstrates how machine learning and NLP techniques can be used for sentiment analysis. TF-IDF helps convert text into numerical form while Naive Bayes efficiently performs classification. The system can predict sentiments for custom user-entered sentences.

Important Viva Questions with Answers

Question	Answer
What is Sentiment Analysis?	It is the process of identifying Positive or Negative emotions from text.
What is NLP?	Natural Language Processing is a field that enables computers to understand
Why is TF-IDF used?	It converts text into numerical values based on word importance.
What is Naive Bayes?	A supervised classification algorithm commonly used for text classification.
What is a Confusion Matrix?	A table used to evaluate prediction performance.
What is Accuracy?	The ratio of correct predictions to total predictions.
What are Stopwords?	Common words like is, the, am that add little meaning.
Why use Machine Learning in sentiment analysis?	It helps automatically classify large amounts of text data.

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